Quiz 7

M 375T Mathematical Foundations of Machine Learning — Fall 2026

15 minutes · closed book · no devices

Name: _____ UT EID: _____

$q_0, \ldots, q_7$ is the Haar basis of $\mathbb{R}^8$, in the order of Practice 7 (constant, then coarse to fine). Let $x = (2, 2, 2, 2, 0, 0, 6, 6)$.

(a) Write $x = \sum_{k=0}^{7} c_k\, q_k$: find the eight numbers c_k. [5 pts]

(b) Write the projection of x onto the span of q_0, q_1 and the projection onto the span of q_0, q_1, q_2, q_3. What is the error energy $\|x - y\|^2$ of each? [5 pts]

Answer:

Total: 10 points. Lowest two quiz scores are dropped at the end of the semester.